

Classification of Motor Imagery EEG Patterns and Their Topographic Representation

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Abstract—We have developed a single trial motor imagery (MI) classification strategy for the brain computer interface (BCI) applications by using time-frequency synthesis approach to accommodate the individual difference, and using the spatial patterns derived from EEG rhythmic components as the feature description. The EEGs are decomposed into a series of frequency bands, and the instantaneous power is represented by the envelop of oscillatory activity, which formed the spatial patterns for a given electrode montage at a time-frequency grid. Time-frequency weights determined by training process were used to synthesize the contributions at the time-frequency domains. The overall classification accuracies for three selected human subjects performing left or right hand movement imagery tasks, were about 87 percent in the ten-fold cross validation without rejecting trials. The loci of motor imagery activity were shown in the spatial topography of differential mode patterns over the sensorimotor area. The present method promises to provide a useful alternative as a general purpose classification procedure for motor imagery classification.

Keywords—Brain computer interface, Motor Imagery, event-related desynchronization, Spatial correlation

I. INTRODUCTION

There is an increasing interest in finding new communication channels between human and environment instead of using nerves and muscles (termed as brain-computer interface, BCI) [1]. Synchronous motor imagery (MI) is one of the major protocols used in the field of BCI. It has been found that planning and execution of movement leads to a short-lasting and circumscribed attenuation in the *mu* (8-12 Hz) and the central *beta* (13-28 Hz) rhythm known as event-related (de-)synchronization (ERD/ERS) which has played an important role in BCI study. EEG rhythmic component related to MI is time-locked to the preparation and execution events and localized asymmetrically over bilateral hemisphere which can be used to classify brain states related to the imagination of limb movements [2]. Usually, subject specific parameters must be set *a priori* without objective criterion. In MI-based BCI system, it is crucial to optimize these sensitive parameters in terms of time, frequency and spatial area to accommodate subject variability. A mental states classification algorithm universally applying to human subjects should be able to set parameters adaptively through training. Positive contributions could come from a large scope of EEG signals with different extent in time, frequency and spatial domain; however, noise artifacts would also deteriorate the

performance if without a proper selecting mechanism. The previous results based on this philosophy have shown the promise only in cases with a small number of channels due to the drawback of the poor generalization capability for involving more channels [3]. Multi-channel based approach using optimal variable-reduction techniques in spatio-temporal combined data has been investigated with favorable results [4]. For lower dimensional features, linear classifiers usually serve well [3, 5] because approximately Gaussian distributions are reasonable expectation in these cases. From these considerations, we present a method to recognize mental states of subjects who are performing synchronous MI tasks (imagination of left or right hand movement) by taking advantage of combining rhythmic components in temporal, frequency and spatial domains. Similarity of patterns measured by correlation coefficients is adopted as a simple category assigning criterion. Weighted synthesizing of spatial patterns over time and frequency bins is then used to augment the performance.

II. METHODOLOGY

A Dataset Description

Scalp EEGs were recorded from fifty-nine channels (placed according to the international 10/20 system) with a 100 Hz sampling rate. The subjects were asked to imagine either right or left hand movement indicated by a highly predictable timed cue on a screen. The timing course of one trial is a six-second-long epoch consisting of several pieces of timing arrangement. At 3.75 s the *preparation cue* lasting 250 ms appeared in a form of English letter “L” or “R” indicating right or left hand movement should be imagined; and at 5.0 s an *execution cue* lasting 50 ms indicated that it was time to make the requested response. For each subject, 180 trials were obtained, 90 for imaginary right hand movement and the other 90 for left. Refer to [6] for details.

B Spatial Filtering and Frequency Decomposition

The raw scalp EEGs are spatially smeared due to the head volume conductor effect. The surface Laplacian can be considered as a spatial high pass filter [7], which is necessary before further signal processing and feature extraction. Here the finite difference implementation of the surface Laplacian was used, with the assumption that the distances from the channel of interest to its neighboring channels are approximately equal [8].

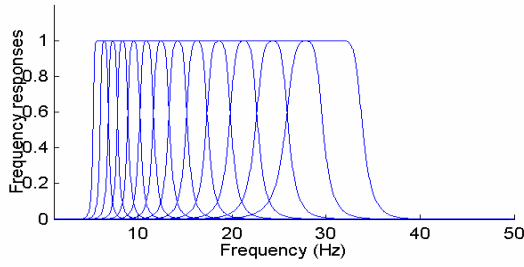


Fig. 1. Frequency bins arrangement by constant-Q methods. Center frequencies are located in the range of 6-30 Hz.

EEG singles were then decomposed into 13 overlapped frequency bins by band-pass filters. We chose constant-Q decomposition scheme. The frequency response shows in Fig. 1. The total range of frequency bands was chosen from around 6 to 30 Hz, which covers both *mu* and *beta* band.

C Instantaneous Power Description

EEG signals were decomposed into groups of different frequency components via the previous procedure, yielding a time signal containing a rapid oscillation modulated by amplitude variation, i.e. envelope of a narrow band signal which can be estimated as the modular part of so called analytic signal obtained by Hilbert transform. These envelopes can be regarded as a smooth description of the instantaneous power of the decomposed EEG rhythm for the frequency bin under consideration [3].

Downsampling of the envelopes was accomplished by integrating their values over several consecutive samples. Specifically, we selected a 2250 ms time range starting from

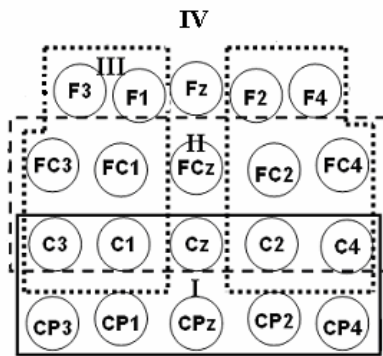


Fig. 2. Four selected montages defined by electrodes within frames numbered from I to III. Montage IV consists of all the 20 electrodes

preparation cue. The new time serial data equivalent to 6.7 Hz sampling rate were obtained by summation of all values within 300 ms time windows with 50% overlapping. All

times mentioned in the context are relative to the raw EEG data.

D Spatial Patterns and Classification

In the training stage, at each time-frequency grid point, we obtained a spatial feature pattern named as *characteristic pattern* at a time-frequency grid, denoted by $P(t, f) = \{P_L(t, f), P_R(t, f)\}$, where the subscript *L* or *R* stands for the events of left or right hand imagined movements. P is a spatial vector consisting of values at each of the channels for an electrode montage. It was calculated by averaging corresponding spatial patterns over all trials in the training dataset. Given an input feature pattern, denoted by $p(t, f)$, the correlation coefficients, $C(t, f) = C(p(t, f), P(t, f))$, between p and P was used to measure which kind of mental state the current trial would be more likely to be. Therefore, we could define an *assignment function* of p as

$$h_{t,f}(p) = \text{sgn}[C(p, P_L) - C(p, P_R)], \quad (1)$$

i.e., $h_{t,f}(p) = 1$ means that p is assigned to left hand movement, and for $h_{t,f} = -1$ for right hand. Go through all trails in training set in this way, the correct classification accuracy denoted by $a_{t,f}$ was obtained.

We hypothesized that these accuracy values reflect the adaptation or sensitivity of the feature patterns at every frequency bin and time segment. We thus defined a normalized *time frequency weight* as

$$w_{t,f} = \begin{cases} [(a_{t,f} - E)/(1 - E)]^m, & a_{t,f} > E \\ 0, & a_{t,f} \leq E \end{cases}, \quad (2)$$

so as to allow $w_{t,f} \in [0, 1]$, where m is a control parameter used to further emphasize those bands with larger accuracy values. E is a threshold which can be set manually.

According to the time-frequency weights, synthetic decision for the whole time segments over all frequency bins was made by the following *synthetic assignment function*

$$g(p) = \text{sgn}[\sum_{t=1}^T \sum_{f=1}^F w_{t,f} h_{t,f}(p)], \quad (3)$$

where T and F denote the sampling numbers in time and frequency domains, respectively. In the testing stage, the classification is accomplished by equation (3). The value of $g(p)$ is either 1 or -1, indicating left and right, respectively. Since $w_{t,f}$ functioned as an indicator of the classification capability for a time segment at a frequency bin, the final classification decision was made by synthesizing all frequency bins and time segments according to their contributions as rated by training.

The weighted sum of characteristic patterns $P(t, f)$ with $w_{t,f}$ at time-frequency grids yields a *synthesized characteristic pattern* (denoted by $\mathbf{P}$), reflecting the features of the event-related potential in time, frequency and spatial domains. Let $\mathbf{P}_L$ and $\mathbf{P}_R$ denote this pattern from left and right MI-EEG, then we shall be able to derive, $\mathbf{P}_L = \mathbf{P}_C + \mathbf{P}_D$, and $\mathbf{P}_R = \mathbf{P}_C - \mathbf{P}_D$, where $\mathbf{P}_C$ is called common-mode pattern (CMP) and $\mathbf{P}_D$ is called differential-mode pattern (DMP).

By simple mathematical manipulation we obtain $\mathbf{P}_C = (\mathbf{P}_L + \mathbf{P}_R)/2$, and $\mathbf{P}_D = (\mathbf{P}_L - \mathbf{P}_R)/2$.

III. RESULTS

A Classified Accuracy

Three subjects out of nine were selected in this study. The results were obtained using 10-fold cross-validation. We set the number of two kinds of EEG patterns to be even in each fold so as to obtain unbiased training performance. After Laplacian filtering of the EEG signals at 59 electrodes, the 10 electrodes at the periphery were excluded from the present analysis. We selected four montages covering different scope of the area of interests for comparing the performance (Fig 2). These electrodes are all selected on the base of the neighboring electrodes of C3 and C4, two most important sites that are close to the sensorimotor area of the cortex.

The average score of three subjects is around 87 percent, and there is only one percentage difference for the different electrode arrangement. For example, the individual scores for montage IV are 83, 91 and 91 in percentage respectively. Others with more or less electrodes or improper locations would deteriorate the performance according to our calculations. Some subjects appear to be more sensitive to montage than others.

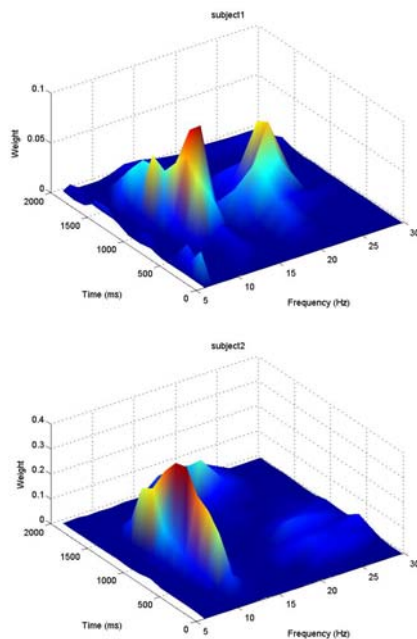


Fig. 3. Time frequency weight distributions of two subjects. Time coordinate was evenly divided into 14 time intervals; zero point indicates the 'preparation cue' and 1250 ms point indicates the 'execution cue'. Frequency coordinate represents 13 center frequencies between 6 and 30 Hz (cf. Fig. 1)

Fig. 1)

B Time-frequency Weights

Time-frequency synthesis plays an important role in enhancing the performance. Because it could increase the classification accuracy through accumulating all the positive contributions designated by training process. Usually at least 5 percent increase achieved comparing with the best score without weighting procedure. The time-frequency-weight distributions for two subjects, as shown in Fig. 3, illustrate the typical performance contribution of characteristic patterns at the time-frequency grids. We could observe that there are peaks mainly located at two (*mu* and *beta*) frequency bands along the time axes, and the magnitudes vary across subjects.

C. Synthesized Spatial Patterns

The synthesized characteristic patterns decomposed into CMP and DMP components are shown in Fig. 4 for two subjects. CMPs usually show symmetrical patterns indicating there are common components occurring for both left and right hand MI tasks, which might reflect the bilateral identity for the hand movement events; and some subjects appeared strong activities in the mid-central area as well. DMPs, on the other hand, explicitly show contralateral attribution of hand movement. Most of sources concentrate in the vicinity of C3 and C4, while others also appear on their neighboring areas.

IV. DISCUSSION

MI tasks would elicit the non-phase-locked EEG components, and can be displayed with different intensity at various frequency bins by frequency decomposition methods. These frequency bins normally lie in the

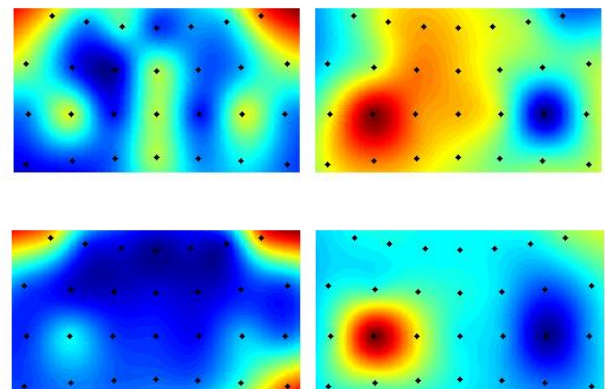


Fig. 4. Topographical mapping of the CMP (left panels) and DMP (right panels) for the two subjects (rows). A clear and consistent contralateral patterns can be observed in DMP maps.

sensorimotor *mu* and *beta* bands, but it is difficult to exactly demarcate these ranges for a given subject. The instantaneous power of the rhythmic components which can be estimated effectively by enveloping distribute over the scalp with distinguishable and unstationary patterns. The present method attempts to employ useful information in time, frequency and spatial domains by accumulating positive contributions to the classification. The human-machine adaptation was implemented in terms of the time-frequency weights evaluated in the training process. The basic classification principle was based on measuring similarity of spatial feature patterns corresponding to each time segment and frequency bin. In addition, the present algorithm is computationally efficient in both training and testing stages, since the major computational procedures are linear weighted summations.

In summary, we have developed a spatio-time-frequency approach for mental state classification based on the rhythmical event related activity during motor imagery tasks from multichannel scalp EEG recordings. This method does not contain subject-dependent parameters, and is computationally efficient. The testing results in three human subjects are promising despite the nature of off-line testing and the fact that no trials are excluded due to noise or artifact. The present method promises to provide a useful alternative as a general purpose classification procedure for MI classification.

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